The mean tau index

$\kappa^\dagger$ as an integer winding; identities first, no unconditional block-mean bound

Stefan Hamann

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Abstract

Write $G^\dagger = (\mathfrak{T}^\dagger)^*\mathfrak{T}^\dagger \succeq 0$ and $\tau^\dagger(s) = \det(I - sG^\dagger)$. All zeros of $\tau^\dagger$ are real and positive, at $s = 1/\lambda_j(G^\dagger)$. Three finite identities hold, with no analytic majorant: (T1) $\kappa^\dagger := \mathrm{ind}_-(R^\dagger - \frac{1}{2}I) = \#\{s \in (0,1) : \tau^\dagger(s) = 0\}$, and $R^\dagger \succeq \frac{1}{2}I$ if and only if $\kappa^\dagger = 0$; (T2) the argument principle on a contour about $(0,1)$; (MI2) the block mean of $\kappa^\dagger$ is the winding of the averaged integrand. The Lean landing site `exists_index_zero_of_block_mean_lt_one` (r430) then turns $\liminf$ of a block mean < 1 into an index-zero window. The honest obstruction to (T2) as a numerical contour is quantified: the 1/2-cluster of $R^\dagger$ maps to zeros of $\tau^\dagger$ in a shrinking collar of $s = 1$. No RH claim.

Status. Lemmas 1–4 and 3 are **SATZ** (finite algebra over $\mathbb{Q}$, machine precision on the named windows). Proposition 1 is a measured anatomy of the contour-approach problem, not a bound. Proposition 2 is a sealed census. The unconditional block-mean bound is *not* claimed (Reviewer-R439). Ausgang `IDENTITIES_EXACT / COLLAR_BOSS_QUANTIFIED / BLOCK_MEAN_CENSUS / UNCONDITIONAL_OPEN`. No L^* claim. No $R^\dagger$ claim. No RH claim.

1 T1: inertia is a zero count

The r409 dictionary is $R^\dagger = (I + G^\dagger)^{-1}$ with $G^\dagger = (\mathfrak{T}^\dagger)^*\mathfrak{T}^\dagger \succeq 0$. Spectral calculus gives

$$R^\dagger - \tfrac{1}{2}I = \frac{I - G^\dagger}{2(I + G^\dagger)}.$$

The factor $2(I+G^\dagger) \succ 0$, so $\mathrm{ind}_-(R^\dagger - \frac{1}{2}I) = \#\{\lambda(G^\dagger) > 1\}$. The zeros of $\tau^\dagger(s) = \det(I - sG^\dagger)$ sit at $s = 1/\lambda_j$, hence in $(0,1)$ precisely when $\lambda_j > 1$.

Lemma 1 (T1). $\kappa^\dagger = \mathrm{ind}_-(R^\dagger - \frac{1}{2}I) = \#\{s \in (0,1) : \tau^\dagger(s) = 0\}$. *In particular* $R^\dagger \succeq \frac{1}{2}I$ *if and only if* $\kappa^\dagger = 0$.

Proof. The displayed congruence, plus $G^\dagger \succeq 0$. Over $\mathbb{Q}$: $G = \mathrm{diag}(2, 1/2)$ gives $R = \mathrm{diag}(1/3, 2/3)$, $R - \frac{1}{2}I = \mathrm{diag}(-1/6, 1/6)$, $\mathrm{ind}_- = 1$, and $\tau(s) = (1 - 2s)(1 - s/2)$ has zeros $\{1/2, 2\}$. The r409 five-atom toy has $C = \left(\begin{smallmatrix} 97 & 23 \\ 23 & 125/16 \end{smallmatrix}\right)$, $\det C = 3661/16$, both eigenvalues of C strictly above 1, hence $\kappa = 0$ and $G = C^{-1}$ at residual $1.2 \cdot 10^{-16}$. On w9 the three counters (eigenvalues of $G^\dagger$ above 1, zeros of $\tau^\dagger$ in $(0,1)$, inertia of $R^\dagger - \frac{1}{2}I$) agree at 0, with dictionary residuals $2 \cdot 10^{-14}$. □

Unaugmented T_0 on w9 still has $\mathrm{nneg}(I - T_0^* T_0) = 1$ (the P1 square). The dagger repairs it: $\|\mathfrak{T}^\dagger\| = 0.99991709 < 1$.

2 T2: argument principle, and the collar

Jacobi/Fredholm variation is the integrand of (T2):

$$\partial_s \log \tau^\dagger(s) = -\operatorname{tr}\big(G^\dagger(I - sG^\dagger)^{-1}\big).$$

Lemma 2 (T2)**.** *For a simple closed contour* Γ *about* $(0,1)$ *that encloses no other zeros of* $\tau^\dagger$*,* $\kappa^\dagger = (1/2\pi i)\oint_\Gamma \partial_s \log \tau^\dagger \, ds$*.*

Proof. Argument principle for a polynomial with only real positive zeros. The integrand identity is the Jacobi formula; a centred finite difference at $s = 0.3$ on w9 matches to relative $4.8 \cdot 10^{-8}$. On the $\mathbb{Q}$ 2×2, a circle of centre $1/2$ and radius 0.49 winds 1 to 10^{-15}. □

The numerical contour on a window is a different question. Eigenvalues of $R^\dagger$ in $[1/2, 3/5]$ map to zeros of $\tau^\dagger$ near $s = 1$ via $s = r/(1-r)$. A circle of radius 0.40 about $1/2$ (the interval $(0.10, 0.90)$) winds 0 on every living window measured: *none* of $\kappa^\dagger$ lives away from $s = 1$. Pushing the radius to 0.499 produces a fractional winding (w9: -1.77), because the vertical side of any rectangle about $(0,1)$ passes within $O(\mathrm{gap}_+)$ of the first exterior zero. For a living window $\mathrm{gap}_+ = 1/\lambda_{\max} - 1$; for a dead window the interior zero sits at $1 - \mathrm{gap}_-$ with gap_- of the same order.

Proposition 1 (Collar anatomy)**.** *On the selected sequence* $k = 3, 4, 5, 6, 7, 9$*,* gap_+ *falls from* $2.98 \cdot 10^{-3}$ *to* $1.91 \cdot 10^{-8}$*. On core-42,* $\mathrm{gap}_+ \in [1.42 \cdot 10^{-7}, 1.66 \cdot 10^{-4}]$ *and* $n_{\mathrm{half}} := \#\{|\lambda(R^\dagger) - 1/2| \le 0.05\}$ *runs* 21 *to* 173*. Dead* χ *place their one interior zero in* $\mathrm{gap}_- \in [4.8 \cdot 10^{-6}, 1.0 \cdot 10^{-3}]$*. Any contour that would count* $\kappa^\dagger$ *must thread this shrinking corridor; that is the boss problem of (T2), not a failure of the identity.*

3 Source form of the integrand

The IIKS/Uvarov layer supplies the characteristic identity $\det(I - sQ) = \det(I - sE)$ and the Uvarov ratio for τ-quotients. Here $Q = G^\dagger$ and the source-side Gram $H = \mathfrak{T}^\dagger(\mathfrak{T}^\dagger)^*$ (a kernel on the X-nodes) has the same non-zero spectrum.

Lemma 3 (Source integrand)**.** $\partial_s \log \tau^\dagger(s) = -\operatorname{tr}\big(H(I - sH)^{-1}\big)$*, equivalently the Dirichlet diagonal* $-\sum_{x \in X}[H(I-sH)^{-1}]_{xx}$*. On the unaugmented interpolant,* $\partial_s \log \tau_0(s) = -\operatorname{tr}\big((K_{d_0}[Y]W_Y - sI)^{-1}\big)$*, where* $K_{d_0}[Y]$ *is the Christoffel–Darboux kernel of the* X*-measure at the hole nodes.*

Proof. Cyclic property of the trace, plus $\det(I_Y - s(\mathfrak{T}^\dagger)^*\mathfrak{T}^\dagger) = \det(I_X - s\mathfrak{T}^\dagger(\mathfrak{T}^\dagger)^*)$. The CD form is $G = D^{-1}K_Y^{-1}D^{-1}$ with $D = W_Y^{1/2}$, checked on w9 at residual $2 \cdot 10^{-12}$; the two traces agree at $2 \cdot 10^{-11}$. Cyclic and slogdet identities for $\mathfrak{T}^\dagger$ hold at $1.4 \cdot 10^{-14}$. □

This is the raw material for a later mean-value theorem on Dirichlet polynomials in the window data. It is not itself a bound.

4 MI2 and the block-mean census

Lemma 4 (MI2)**.** *On any common contour,*

$$\frac{1}{K}\sum_{k=1}^{K} \kappa_k^\dagger = \frac{1}{2\pi i}\oint \partial_s \Big(\frac{1}{K}\sum_{k=1}^{K} \log \tau_k^\dagger\Big)\, ds.$$

Proof. Linearity of the integral. On the $\mathbb{Q}$ pair $\operatorname{diag}(2, 1/2)$ and $\operatorname{diag}(0.4, 0.3)$, a common circle of radius 0.40 gives mean winding 1/2 equal to the averaged-integrand winding. On $\{w9, \mathrm{kz}\,5\}$ the same identity holds at $4{\cdot}10^{-15}$. The Lean arithmetic `exists_index_zero_of_block_mean_lt_one` then yields an index-zero window in any block whose mean is strictly less than 1. $\square$

Proposition 2 (Census). *Selected* $k = 3, 4, 5, 6, 7, 9$*:* $\kappa^\dagger = 0$ *on* $6/6$*, block mean* $0 < 1$ *(*$k = 8$ *not rebuilt; r421 class). Core-42:* $\kappa^\dagger = 0$ *on* $42/42$*, mean* 0*; no growth of* $\kappa^\dagger \geq 1$ *with* k*. Dead* χ $6/6$*:* $\kappa^\dagger = 1$ *exactly (*χ_3-15/19/23/33/39 *and* χ_4-20*). Living* χ_3-9*:* $\kappa^\dagger = 0$.

The selected sequence is entirely living. The empirical mean is far below 1. The honest remaining question is an *unconditional* bound that keeps the mean below 1 on infinitely many blocks — not claimed here.

5 Kills and delimiters

- ***s*-No-Go.** The r265 no-go closed s as a monotone positivity dynamics (the endpoint *was* the budget). Here s only counts zeros. Witness: $\tau^\dagger(1/2) > 0$ on both living w9 and dead χ_3-15 (the κ-zero sits in the collar, not at mid-interval).
- **Soft mutants.** $\operatorname{tr} G^\dagger$ is 50.9 on w9 ($\kappa^\dagger = 0$) and 53.0 on dead-15 ($\kappa^\dagger = 1$). The r398 moment $M_2 = 16 \operatorname{tr}((I - R^\dagger)^4)$ is 32.96 vs 39.01, both $\gg 2$. The 1/2-cluster dominates every even moment; moments do not see the threshold.
- **Scramble.** Unaugmented $\operatorname{nneg}(R) = 21$ (bulk already dead). The dagger is not the story.
- **Circularity gate.** A pointwise bound $|\sum \Lambda n^{-1/2-it}| \ll X^\varepsilon$ is forbidden as a mean-value carrier (it would close the route by assuming the conclusion at the prime-sum scale).

Claim boundary

Nothing in this note is evidence for or against the Riemann hypothesis, in either direction. The identities are finite linear algebra. The census is a finite table. The unconditional block-mean bound is open.

References

[1] Round 265, `s_monotonicity`: s-coordinate class closed (naive restatement; IIKS definite dynamics, wrong endpoint).

[2] Round 398, `high_moment_inertia`: even moments blinded by the 1/2-cluster.

[3] Round 409, `borodin_birkhoff_intertwiner`: $\mathfrak{T}^\dagger$ constructor and $R = (I + G)^{-1}$.

[4] Round 430, `RH/FrequentlySelected.lean`: `exists_index_zero_of_block_mean_lt_one`.

[5] Round 433, `edge_redheffer`: last pivot $\delta = 1 - q^\dagger$.

[6] Reviewer-R439 (not this round): unconditional block mean.